

Supplementary Information

Complete Patient-Level Leakage Among Traceable Images in a Widely Used Brain Tumor MRI Benchmark

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Supplementary Methods

Provenance sensitivity control

The initial provenance search used ImageNet-pretrained ResNet-18 embedding cosine similarity, histogram-equalized pixel correlation, and intersection-over-union of thresholded brain masks. A byte-identical positive control returned 1.0 under every measure but did not test robustness to transformations introduced by an aggregation pipeline. We therefore constructed transformed-positive controls from known figshare slices. JPEG re-encoding at quality 85–95 and resizing preserved cosine similarity above 0.999, whereas cropping to the brain region reduced median equalized correlation to 0.601 and embedding cosine to 0.963. After crop-normalizing both collections and searching all rotations and reflections, transformed true matches reached median cosine 0.9988 and minimum 0.9884. We used a conservative acceptance threshold of 0.98.

The corrected search matched 4,586 of 5,023 tumor images. Of these, 2,956 matched after a 90-degree rotation and 1,630 after reflection plus a 90-degree rotation; none matched in the identity orientation. Three aggregated images labelled pituitary were labelled meningioma in figshare. The released `figshare_patient_map.json` preserves every accepted score and orientation.

Split checks

Each patient-disjoint fold was generated from a single cached five-fold assignment. Each recovered patient appears in exactly one test fold. The unmatched tumor images are training-only, and no-tumor near-duplicate clusters are indivisible. Every materialized fold asserts zero group overlap and zero pixel-identical images across the training/test boundary. The random control likewise uses one cached partition rather than five independent draws; every eligible image appears in one test fold, and per-class fold sizes match the patient arm exactly.

Supplementary Tables

Table S1: Composition of the released folders.

Class	Training	Testing	Total
Glioma	1,321	300	1,621
Meningioma	1,339	306	1,645
No tumor	1,595	405	2,000
Pituitary	1,457	300	1,757
Total	5,712	1,311	7,023

Table S2: Crop-normalized, orientation-searched matching against figshare.

Quantity	Value
Transformed control, median cosine	0.9988
Transformed control, minimum cosine	0.9884
Acceptance threshold	0.98
Tumor images matched	4,586/5,023 (91.3%)
Figshare patients recovered	233/233
Matched after rotation	2,956
Matched after reflection and rotation	1,630
Label disagreements	3

Table S3: Exact duplication and within-dataset redundancy by class. “Unique” counts distinct decoded pixel contents; clusters additionally collapse embedding cosine similarity ≥ 0.99 .

Class	Test duplicates	% of test class	Files	Unique	Clusters
Glioma	0	0.0%	1,621	1,620	1,620
Meningioma	106	34.6%	1,645	1,531	1,522
No tumor	105	25.9%	2,000	1,706	1,113
Pituitary	5	1.7%	1,757	1,740	1,739
Total	216	16.5%	7,023	6,597	5,994

Table S4: Training hyperparameters used for all reported primary runs.

Setting	Value
Architecture	ImageNet-pretrained ResNet-18
Input	224×224 , grayscale replicated to RGB
Optimizer	Adam
Learning rate	10^{-4}
Weight decay	0 (Adam default)
Batch size	32
Maximum epochs	30
Early stopping patience	5 epochs
Scheduler	ReduceLROnPlateau, factor 0.5, patience 2
Augmentation	Flip; $\pm 10^\circ$ rotation; 10% brightness/contrast
Primary seeds	0, 1, 42 per fold

Table S5: Analyses corrected or superseded during this work. Every result in the submitted manuscript reflects the right-hand column.

Analysis	Earlier problem	Submitted analysis
Provenance	An identical-file positive control did not test robustness to cropping or orientation changes, producing a false-negative source trace.	Transformed-positive controls motivated crop-normalized, orientation-aware matching, which recovered 4,586 patient identifiers.

Analysis	Earlier problem	Submitted analysis
Validation split	The initial train/validation division was not grouped by patient, contaminating checkpoint selection, early stopping, and temperature fitting.	All cross-validation runs use group-disjoint validation by recovered patient or no-tumor near-duplicate cluster.
Duplicate errors	A Fisher test treated predictions from ten models on the same images as independent.	Model-level paired and image-level permutation tests preserve correlation; neither reached conventional significance.
Deduplication bound	A bound on inflation was inferred from a p -value.	The reported 95% interval for the accuracy difference is -0.03 to +0.43 points.
Variance claim	Seed variance was compared with an inapplicable binomial floor.	That claim was withdrawn and replaced by per-image agreement and fold-by-seed variance analyses.
Patient estimate	A single partition gave $94.75\% \pm 0.80$, where the interval reflected seeds but held the partition fixed.	Five-fold patient-disjoint cross-validation gives 95.19%; partition variation (2.18 points) exceeds seed variation (0.83 points).
Paired comparison	The initial 1,061-image comparison included 300 no-tumor images without patient identifiers.	Restriction to 761 images with recovered identifiers gives 99.12% versus 93.60% and a 7.3-fold error-rate ratio.
Random control	Independently generated folds did not form one matched partition of the evaluation population.	One cached five-fold random partition matches patient-fold class counts and covers the same 6,292 images exactly once per seed.

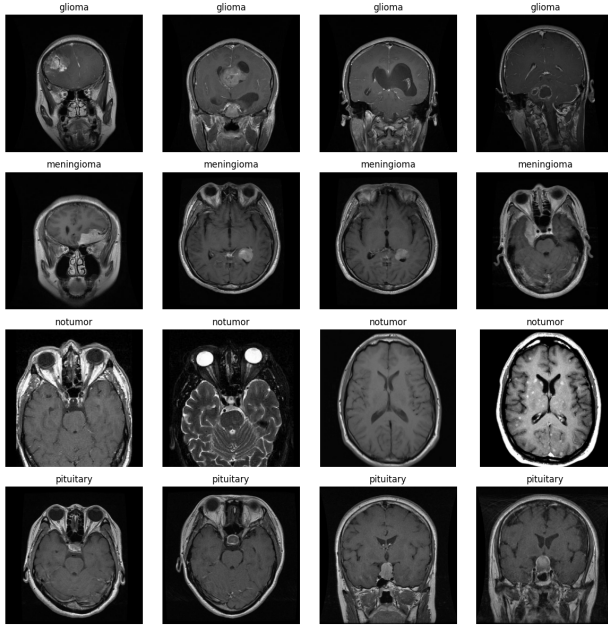
Table S6: Released-split baseline and ablation results. The main configuration uses ten seeds; the other ablation cells use five.

Augmentation	Backbone	Accuracy	SD
On	Fully fine-tuned	99.22%	0.16 pt
Off	Fully fine-tuned	98.66%	0.47 pt
On	Frozen	83.57%	0.42 pt
Off	Frozen	84.99%	0.31 pt

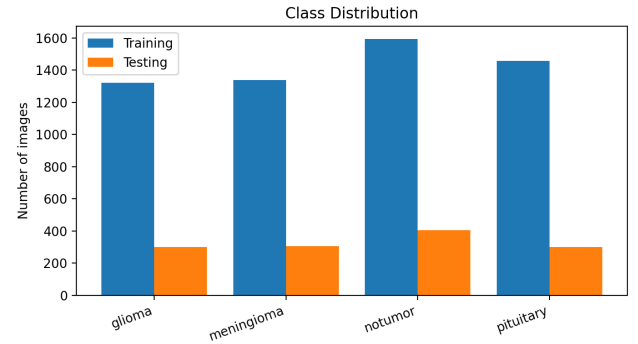
Table S7: Complete run inventory. Prediction files and per-run histories are available under `runs/`; aggregate metrics are stored in each tagged directory.

Tag	Configuration	Evaluation	n
leaky	Augmented, full fine-tuning	Released holdout	10
dedup	Augmented, full fine-tuning	Deduplicated holdout	10
cv_fold0--4	Augmented, grouped validation	Patient-disjoint CV	15
rand_fold0--4	Augmented, grouped validation	Random-fold control	15
patient	Augmented, ungrouped validation	Retired single partition	5
abl_noaug_full	No augmentation, full fine-tuning	Released holdout	5
abl_aug_frozen	Augmented, frozen backbone	Released holdout	5
abl_noaug_frozen	No augmentation, frozen backbone	Released holdout	5
Total			70

Supplementary Figures

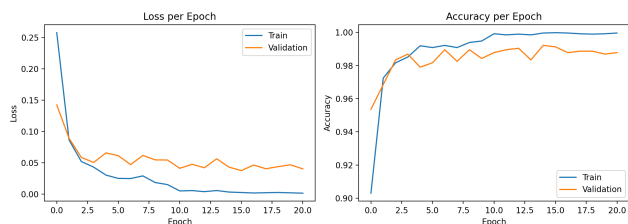


(a) One example from each class.

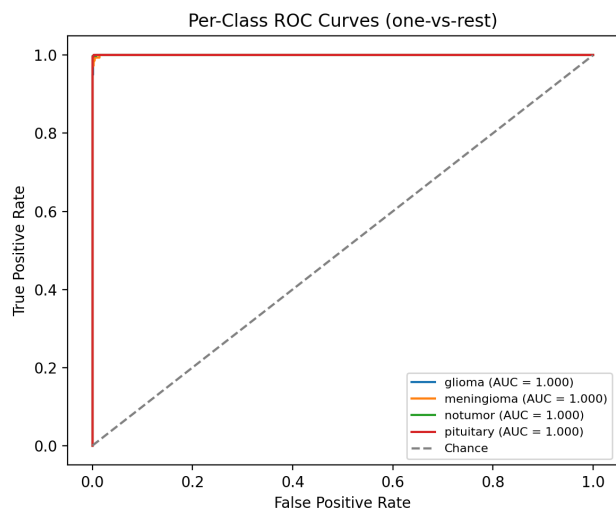


(b) Released folder counts by class.

Figure S1: Dataset examples and class distribution.

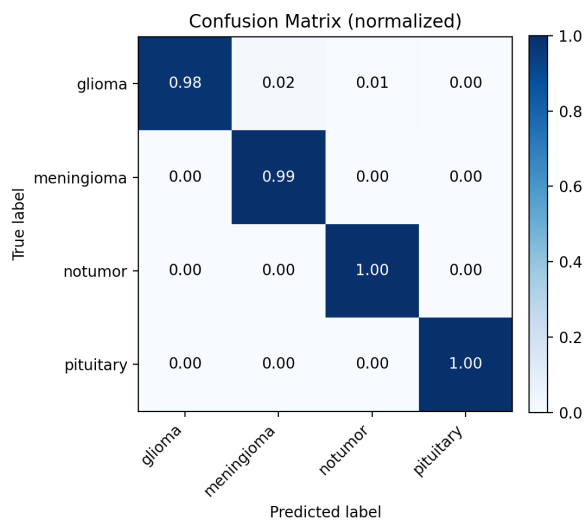


(a) Training and validation loss and accuracy.

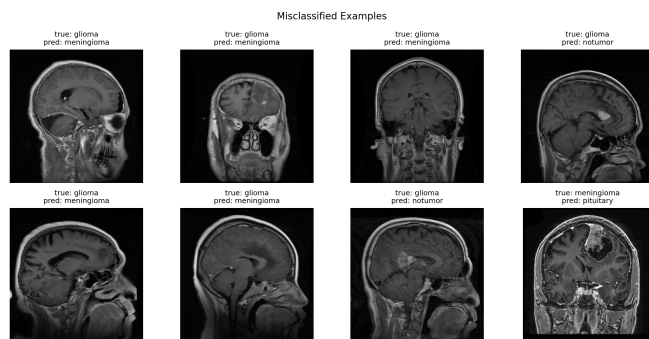


(b) Released-split one-versus-rest ROC curves.

Figure S2: Released-split training dynamics and ranking performance. These results are measured under patient leakage.



(a) Normalized confusion matrix.



(b) Released-split errors.

Figure S3: Released-split error analysis, provided for reproducibility rather than as evidence of generalization.